

Scalable Design and Optimization for Frequency-Domain Wave and Vibration Problems

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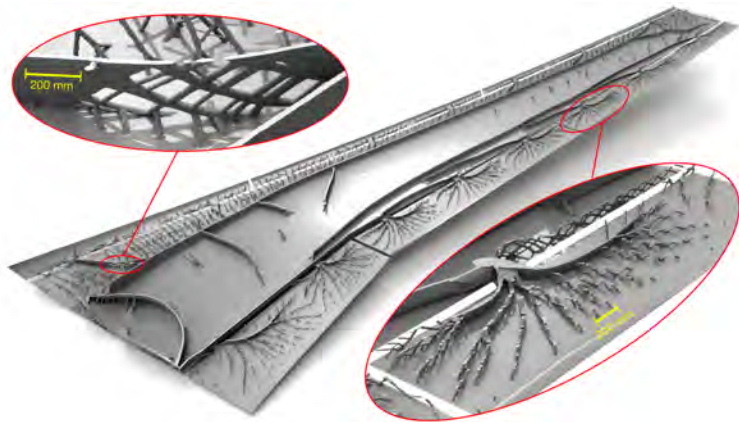
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Minimize weight/maximize stiffness



Airbus A380

Topology optimization in aerospace



Topology optimization

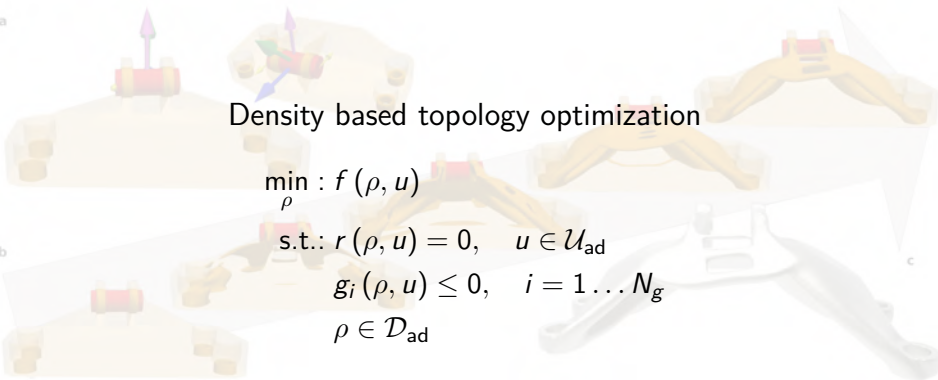
Density based topology optimization

$$\min_{\rho} : f(\rho, u)$$

$$\text{s.t.: } r(\rho, u) = 0, \quad u \in \mathcal{U}_{\text{ad}}$$

$$g_i(\rho, u) \leq 0, \quad i = 1 \dots N_g$$

$$\rho \in \mathcal{D}_{\text{ad}}$$



Vibration analysis and optimization

Fatigue



Imprecise fixture



Resonant Ultrasound Spectroscopy



Analysis in time and frequency domains

- Time domain discretized system:

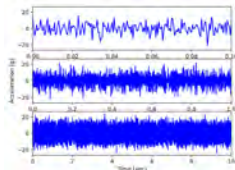
$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{I}(t)$$

- $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{n \times n}$, $\mathbf{K}, \mathbf{M} \succ \mathbf{0}$,
 $\mathbf{C} \succeq \mathbf{0}$
- Harmonic excitation - $\mathbf{I}(t) = \mathbf{f}e^{i\omega t}$,
response $\mathbf{u} = \mathbf{q}e^{i\omega t}$
- Frequency domain system:

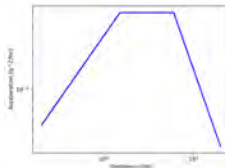
$$(-\omega^2 \mathbf{M} + i\omega \mathbf{C} + \mathbf{K}) \mathbf{q} = \mathbf{f}$$

- Limited to linear elastic material behavior

Excitation time history



Power spectral density



Scalar and vector time-harmonic models

- Start from time-domain *isotropic* linear elasticity

$$\rho_m(\rho)\ddot{\mathbf{u}} + \mathcal{D}_\rho\dot{\mathbf{u}} - \nabla \cdot \boldsymbol{\sigma}_\rho(\mathbf{u}) = \mathbf{l}(\mathbf{x}, t)$$

- For acoustics, use the separate scalar pressure Helmholtz model , extended here with first-order radiation boundaries and a design-independent sponge layer
- Both models share the same frequency-domain polynomial

$$\left(\mathbf{K}_\chi(\rho) - \omega^2 \mathbf{M}_\chi(\rho) + i\omega \mathbf{C}_\chi(\rho) \right) \mathbf{q}_\chi = \mathbf{f}_\chi, \quad \chi \in \{p, e\}$$

- $\mathbf{K}_\chi, \mathbf{M}_\chi \succ \mathbf{0}$, $\mathbf{C}_\chi \succeq \mathbf{0}$; elasticity uses Rayleigh damping
 $\mathbf{C}_e = \alpha \mathbf{M}_e + \beta \mathbf{K}_e$
- One solver framework covers scalar acoustics *and* vector elastodynamics

Optimization problem and real-block forward/adjoint

- Band-aggregated objective $J(\rho) = \sum_k w_k j(\rho, \mathbf{q}_r(\omega_k), \mathbf{q}_i(\omega_k))$;
volume constraint $\frac{1}{|\Omega|} \int_{\Omega} \tilde{\rho} dx \leq v^*$
- Real, non-symmetric block state equation

$$\mathbf{A}(\omega) \begin{bmatrix} \mathbf{q}_r \\ \mathbf{q}_i \end{bmatrix} = \begin{bmatrix} \mathbf{f}_r \\ \mathbf{f}_i \end{bmatrix}, \quad \mathbf{A}(\omega) = \begin{bmatrix} \mathbf{W} & -\mathbf{T} \\ \mathbf{T} & \mathbf{W} \end{bmatrix}$$

with $\mathbf{W} = \mathbf{K} - \omega^2 \mathbf{M}$ and $\mathbf{T} = \omega \mathbf{C}$

- Adjoint uses $\mathbf{A}(\omega)^T$ – same $\mathbf{W}$, off-diagonal $\mathbf{T}$ up to sign
- $\Rightarrow$ forward and adjoint at a fixed ω **reuse the same preconditioner setup**

Why the frequency-domain equations are hard to solve

- $\mathbf{A}(\omega) = \mathbf{K} - \omega^2 \mathbf{M} + i\omega \mathbf{C}$ – **complex**, and **indefinite** once $\omega^2 > \lambda_1$; near resonance $\mathbf{W}(\omega) = \mathbf{K} - \omega^2 \mathbf{M}$ is nearly singular
- Operator changes at **every design update** and **every frequency**, with forward *and* adjoint RHS $\Rightarrow$ repeated linear solves dominate wall-clock time

Direct solvers	Iterative / Krylov solvers
robust, black-box, accuracy independent of ω	low memory, matrix-free, GPU-friendly, reuse across RHS
3D fill-in: memory / time wall; new factorization per ω	plain Krylov stalls for indefinite / near-resonance systems
here: reference + inner half-step solves only	needs a problem-specific preconditioner valid across crossings

PRESB block preconditioning $\rightarrow$ indefinite splitting $\rightarrow$ spectral two-level correction

Modal reduction and its limits

- Generalized eigenproblem $(\mathbf{K} - \lambda_i \mathbf{M}) \psi_i = \mathbf{0}$, $\Psi^T \mathbf{K} \Psi = \mathbf{\Lambda}$, $\Psi^T \mathbf{M} \Psi = \mathbf{I}$
- Rayleigh damping $\Rightarrow \Sigma = \Psi^T \mathbf{C} \Psi$ diagonal; approximate $\mathbf{q} \approx \Psi \mathbf{p}$ with

$$\left(-\omega^2 \mathbf{I} + i\omega \Sigma + \mathbf{\Lambda} \right) \mathbf{p} = \Psi^T \mathbf{f}$$

- Cheap, near-diagonal solve for every ω – *once* Ψ is available
- **But:** accuracy tied to mode count m ; localized / high-derivative adjoints leak outside $\text{span}(\Psi)$; broadband objectives need m to grow; Ψ must be recomputed as the design changes
- Used here only as a **baseline**; selected modes reappear later as a *coarse correction*, not a replacement for the full solve

Optimization framework and building blocks

- **Design space / filters** – PDE filter, projection; guarantees a well-posed problem
- **Forward + adjoint solvers** – the frequency-domain systems of this talk; every objective / constraint is a reusable class supplying its adjoint right-hand side and gradient
- **Design-update algorithms:**
 - OC – single constraint
 - **Native MMA in MFEM** – parallel, no external optimizer dependency, multiple non-linear constraints
 - **SiMPL** – Sigmoidal Mirror descent with a Projected Latent variable; Fermi–Dirac entropy / Bregman divergence; pointwise bound-preserving even at high order; mesh-independent convergence [Keith et al., 2025, Kim et al., 2025]
- All components share MFEM's operator / preconditioner interfaces

The PRESB preconditioner

- Complex system $(\mathbf{W} + i\mathbf{T})\mathbf{x} = \mathbf{b}$, $\mathbf{W} = \mathbf{K} - \omega^2\mathbf{M}$, $\mathbf{T} = \omega\mathbf{C}$, as a real block $\mathbf{A} = \begin{bmatrix} \mathbf{W} & -\mathbf{T} \\ \mathbf{T} & \mathbf{W} \end{bmatrix}$ acting on $[\mathbf{x}_r; \mathbf{x}_i]$
- Preconditioner $\mathbf{P} = \begin{bmatrix} \mathbf{W} + 2\mathbf{T} & -\mathbf{T} \\ \mathbf{T} & \mathbf{W} \end{bmatrix}$; for $\mathbf{W}, \mathbf{T} \succeq \mathbf{0}$, $\mathbf{W} + \mathbf{T} \succ \mathbf{0}$:
 $\frac{1}{2} \leq \lambda(\mathbf{P}^{-1}\mathbf{A}) \leq 1$
[Axelsson and Salkuyeh, 2019, Axelsson and Karátson, 2020]
- $\mathbf{P}^{-1}$ applied = two solves with the same SPD matrix $\mathbf{W} + \mathbf{T}$ (one factorization / AMG setup per application):
 - 1) $(\mathbf{W} + \mathbf{T})(\mathbf{x}_r - \mathbf{x}_i) = \mathbf{b}_r - \mathbf{b}_i$
 - 2) $(\mathbf{W} + \mathbf{T})\mathbf{x}_i = \mathbf{b}_i - \mathbf{T}(\mathbf{x}_r - \mathbf{x}_i)$, 3) $\mathbf{x}_r = (\mathbf{x}_r - \mathbf{x}_i) + \mathbf{x}_i$
- **Limitation:** $\mathbf{W} \succ \mathbf{0}$ only for $\omega^2 \leq \lambda_1$ – indefinite past the first resonance

Block-diagonal alternative and MINRES

[Axelsson and Karátson, 2020]

- Set $\mathbf{y} := -\mathbf{x}_i \Rightarrow$ symmetric-indefinite real system

$$\begin{bmatrix} \mathbf{W} & \mathbf{T} \\ \mathbf{T} & -\mathbf{W} \end{bmatrix} \begin{bmatrix} \mathbf{x}_r \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_r \\ \mathbf{b}_i \end{bmatrix}$$

- Block-diagonal preconditioner $\mathbf{P}_{\text{BD}} = \text{diag}(\mathbf{H}, \mathbf{H})$,
 $\mathbf{H} = \mathbf{W} + \mathbf{T}$ – two independent solves with the *same* $\mathbf{H}$ as PRESB
- If $\mathbf{W}, \mathbf{T} \succeq \mathbf{0}$, $\mathbf{H} \succ \mathbf{0}$: $\sigma(\mathbf{P}_{\text{BD}}^{-1} \mathbf{A}_s) \subset [-1, -\frac{1}{\sqrt{2}}] \cup [\frac{1}{\sqrt{2}}, 1]$
- Symmetric $\Rightarrow$ MINRES: short recurrence, fixed Krylov-vector memory (vs. growing Arnoldi basis in GMRES)
- Attractive when Krylov-vector memory is the constraint; both preconditioners share the existing solver interface

Applicability of the unsplit preconditioners

- $W(\omega) = K - \omega^2 M$, $T(\omega) = \omega C$, $H(\omega) = K - \omega^2 M + \omega C$
- Uniform spectral bounds require $\omega^2 \leq \lambda_1$ and $H \succ 0$
- Rayleigh damping $C = \alpha M + \beta K$:

$$H \succ 0 \iff (1 + \omega\beta) \lambda_1 + \omega\alpha - \omega^2 > 0$$

- Damping buys a *finite* interval past the first crossing – but **no frequency-uniform clustering** there
- Once $H \not\succ 0$: SPD inner actions in PRESB and the SPD preconditioner required by MINRES are both lost
- $\Rightarrow$ neither unsplit method is a stand-alone choice for a broadband resonance-crossing sweep

Unsplit performance – exact H^{-1}

- Cantilever benchmark, $\Omega_2 = (0, 4) \times (0, 1)$,
 $\Omega_3 = (0, 4) \times (0, 1)^2$; $p = 1, 2, 3$; MUMPS applies H^{-1}

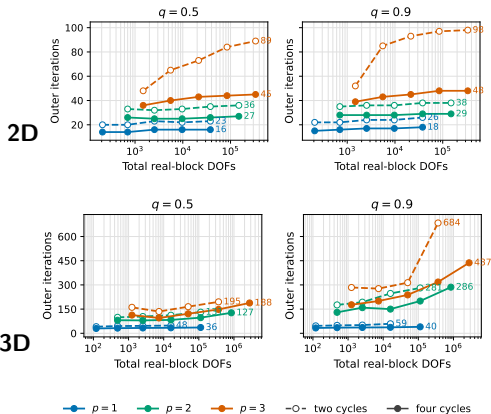
Prec.	Solver	Spectral enclosure (H^{-1} exact)	Its.
PRESB	GMRES	$[1/2, 1]$	4
Block diag.	MINRES	$[-1, -1/\sqrt{2}] \cup [1/\sqrt{2}, 1]$	7

- Counts **independent** of mesh size, dimension and polynomial order
- MINRES needs $\approx 2 \times$ the polynomial degree (two-sided symmetric enclosure) $\Rightarrow 7$ vs. 4

Unsplit performance – approximate LOR-AMG inverse

Outer iterations vs. mesh refinement; $q = \omega/\omega_1$; exact $\mathbf{H}^{-1}$ gave 4 / 7.

2D flat in mesh; 3D flat only for $p = 1$ – high-order needs a p -robust inner solve.



Indefinite splitting – Method I

- **Idea:** split off the indefinite part so PRESB always sees an SPSP real block: $\mathbf{W} = \mathbf{W}_1 - \mathbf{W}_2$, $\mathbf{W}_1 = \mathbf{K}$, $\mathbf{W}_2 = \omega^2 \mathbf{M} (\succ \mathbf{0}, \omega > 0)$, $\mathbf{T} = \omega \mathbf{C} \succeq \mathbf{0}$ [Axelsson et al., 2022]
- Stationary two-half-step iteration, each half-step = one PRESB solve:

$$\begin{aligned}(\mathbf{W}_1 + i\mathbf{T}) \mathbf{x}^{(j+1/2)} &= \mathbf{W}_2 \mathbf{x}^{(j)} + \mathbf{b} \\ (\mathbf{W}_2 - i\mathbf{T}) \mathbf{x}^{(j+1)} &= \mathbf{W}_1 \mathbf{x}^{(j+1/2)} - \mathbf{b}\end{aligned}$$

- Fixed-point form $\mathbf{x}^{(j+1)} = \mathbf{G} \mathbf{x}^{(j)} + \mathbf{c}$, convergent $\iff \rho(\mathbf{G}) < 1$, with $\mathbf{G} = (\mathbf{W}_2 - i\mathbf{T})^{-1} \mathbf{W}_1 (\mathbf{W}_1 + i\mathbf{T})^{-1} \mathbf{W}_2$
- One sweep from $\mathbf{x}^{(0)} = \mathbf{0}$ = the preconditioner action $\mathcal{S}_\omega^{-1} \approx \mathbf{A}^{-1}$ for the *unmodified* GMRES

Indefinite splitting – Method III

[Axelsson et al., 2022]

- Parameterized split, $\eta \geq 1$

$$1) (\eta \mathbf{T} + i\mathbf{W}_2)\mathbf{x}^{(j+1/2)} = ((\eta - 1)\mathbf{T} + i\mathbf{W}_1)\mathbf{x}^{(j)} - i\mathbf{b}$$

$$2) (\eta \mathbf{T} - i\mathbf{W}_1)\mathbf{x}^{(j+1)} = ((\eta - 1)\mathbf{T} - i\mathbf{W}_2)\mathbf{x}^{(j+1/2)} - i\mathbf{b}$$

- Half-steps use PRESB with block pairs $(\eta \mathbf{T}, \mathbf{W}_2)$ and $(\eta \mathbf{T}, \mathbf{W}_1)$
- Mapping to [Axelsson et al., 2022]:

this talk	Axelsson et al. 2022
Method I	Method I, Eq. (5)
Method III	parameterized Method III, Eq. (9), $\alpha \rightarrow \eta$

- Every Method-III result reports η and its selection cost; η is never tuned on the reference solution

Convergence near eigenvalue crossings

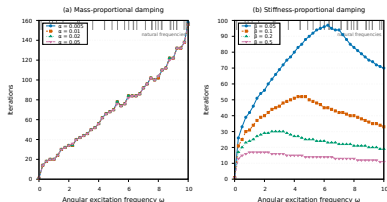
- Rayleigh damping $\Rightarrow \mathbf{K}, \mathbf{M}, \mathbf{C}$ share a modal basis; analyze one mode at a time
- One Method-I sweep multiplies the error in mode i by g_i , with

$$|g_i| = \frac{a_i}{\sqrt{1 + a_i^2}} \frac{b_i}{\sqrt{1 + b_i^2}} < 1$$

- With $d_i = \alpha + \beta \lambda_i$: $a_i = \frac{\lambda_i}{\omega d_i}$ (stiffness/damping), $b_i = \frac{\omega}{d_i}$ (inertia/damping)
- Fast reduction if *either* real block is small relative to damping; slow only when *both* ratios are large
- $\rho = \max_i |g_i|$ – worst-mode error fraction per stationary sweep

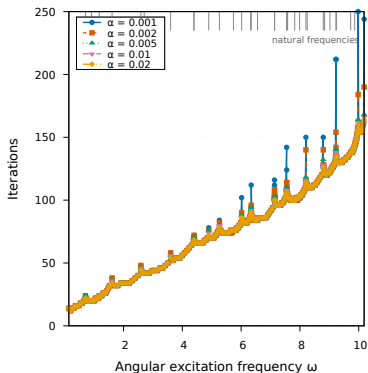
Mass- vs. stiffness-proportional damping

- Mass-proportional $\mathbf{C} = \alpha \mathbf{M}$:
 $\zeta_i \propto 1/\omega_i$ – high modes lightly damped, **sharp** resonance
iteration peaks
- Stiffness-proportional $\mathbf{C} = \beta \mathbf{K}$:
 $\zeta_i \propto \omega_i$ – high modes strongly / over-damped, peaks broadened or absent
- At resonance $\omega^2 = \lambda_i$:
closed-form contraction factor;
Method III has a **resonance-optimal** η

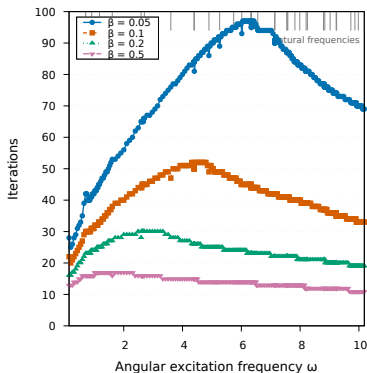


Splitting experiments – elastic beam, Method I

(a) Mass-proportional damping

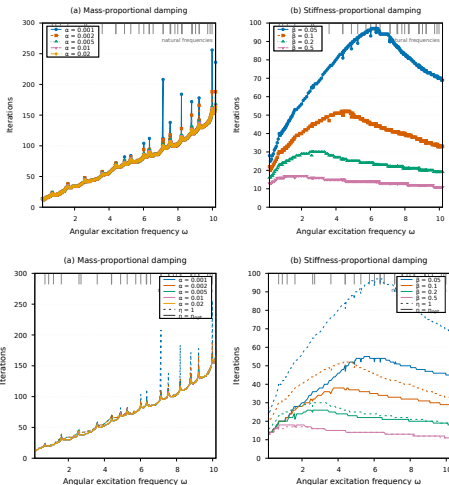


(b) Stiffness-proportional damping



Outer GMRES iterations, S_{ω}^{-1} alone; gray ticks = natural frequencies. **Resonance peaks** kept by mass damping (left), removed by stiffness damping (right).

Splitting experiments – Method III



Method III: $\eta = 1$ (top) vs. resonance-optimal η (bottom).

Limitations of the single-level splitting smoother

- (L1) Near-resonance degradation – $\mathbf{W}(\omega_k)$ nearly singular as $\omega_k^2 \rightarrow \lambda_i$
- (L2) No reuse across the sweep – every ω_k solved from a cold start
- (L3) Repeated setup cost – half-step operators change with ω
- (L4) Restricted closed-form theory – modal factor assumes Rayleigh damping
- (L5) Degradation with polynomial order – AMG on high-order SPD blocks loses effectiveness as p grows

Spectral two-level preconditioner addresses (L1) and partly (L2); LOR preconditioning addresses (L5); (L3) is measured separately; (L4) is a scope caveat.

Polynomial-order-robust preconditioning via LOR

- Low-order-refined (LOR) preconditioning for limitation (L5)
- Auxiliary low-order FE matrix on a mesh refined to the high-order nodal points
- Spectrally equivalent to the high-order SPD block,
equivalence constant independent of p
- Passed to standard AMG in place of the dense assembled high-order matrix
- A fixed number of AMG-on-LOR V-cycles = the fixed linear inner inverse inside the splitting
- Built on MFEM matrix-free + LOR infrastructure, with CPU and **GPU** backends
[Anderson et al., 2021, Pazner, 2020, Pazner et al., 2023]

Spectral two-level preconditioner – coarse space

- Generalized eigenvectors $\mathbf{K}\psi_i = \lambda_i \mathbf{M}\psi_i$ with $\sqrt{\lambda_i}$ in the target band $[\omega_{\min}, \omega_{\max}]$ (plus guard δ_ω)
- Real-block coarse basis

$$\mathbf{Z} = \begin{bmatrix} \Psi_c & \mathbf{0} \\ \mathbf{0} & \Psi_c \end{bmatrix} \in \mathbb{R}^{2n \times 2q}$$

– each physical mode represents both real and imaginary components

- **Not modal reduction:** GMRES still solves the full $2n$ system; the coarse solve only corrects the selected error components
- Same constrained FE space for eigenproblem and driven problem; eliminate essential DOFs *after* forming the block operator

Galerkin correction and splitting smoother

- Coarse operator and solve at frequency ω

$$\mathbf{E}(\omega) = \mathbf{Z}^T \mathbf{A}(\omega) \mathbf{Z}, \quad \mathcal{C}_\omega^{-1} = \mathbf{Z} \mathbf{E}(\omega)^\dagger \mathbf{Z}^T$$

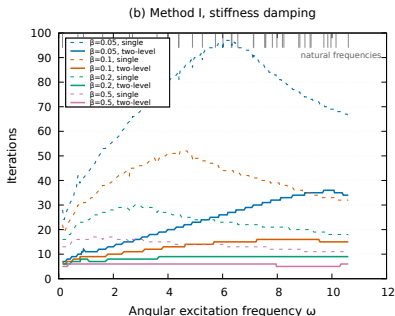
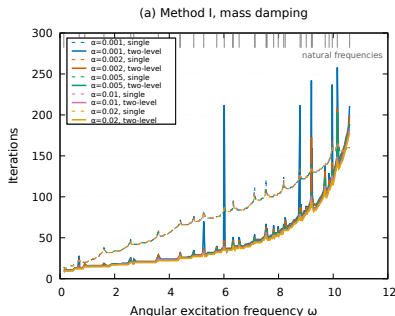
(dense SVD; effective rank reported)

- Multiplicative two-level preconditioner, one pre-smoothing sweep

$$\mathcal{B}_{2L,\omega}^{-1} = \mathcal{S}_\omega^{-1} + \mathcal{C}_\omega^{-1} \left(\mathcal{I} - \mathbf{A}(\omega) \mathcal{S}_\omega^{-1} \right)$$

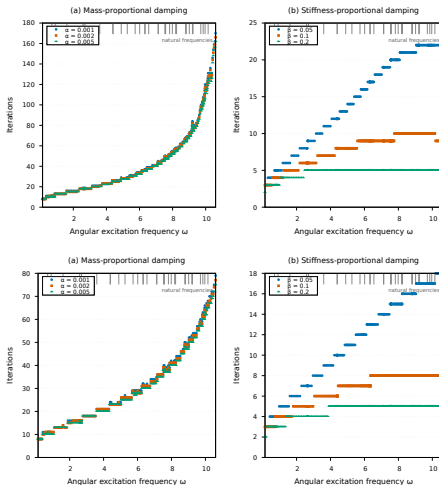
- Modal 2×2 block $\begin{bmatrix} \lambda_i - \omega^2 & -\omega(\alpha + \beta \lambda_i) \\ \omega(\alpha + \beta \lambda_i) & \lambda_i - \omega^2 \end{bmatrix}$ – nonsingular at $\omega^2 = \lambda_i$ when the mode is damped
- Exposed through the same Apply; adjoint reuses $\mathbf{E}_a = \mathbf{E}^T$ (transposed coarse solve)

Two-level results – elastic beam (Method I)



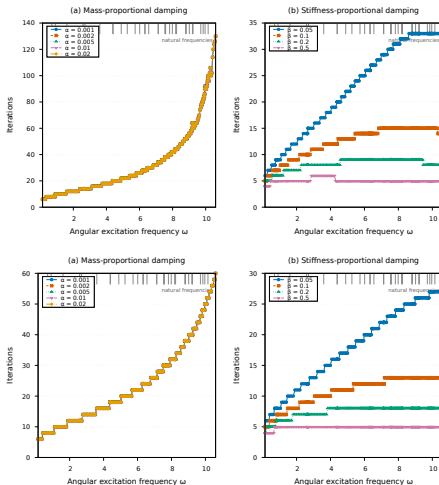
Single-level (dashed) vs. spectral two-level (solid), 36-mode coarse space (real-block rank 72), refined beam (37,828 real-block unknowns). Mean outer count **68.4** $\rightarrow$ **37.1** (-45.8% over 3069 pairs); every stiffness-damped case improved (max 97, 52, 30, 17 $\rightarrow$ 36, 16, 9, 6). Mean solve time 3.52 $\rightarrow$ 1.85 s.

Smoothing and coarse-space size sensitivity



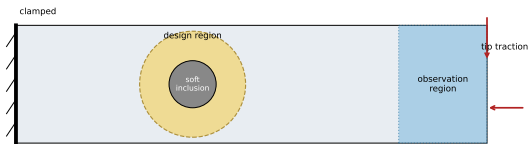
Forward pre+post smoothing: 36 modes (top), 50 modes (bottom); mean $36.9 \rightarrow 21.9$.

Projection-first deflation alternative



Projection-first deflated Method I: 36 (top) / 50 (bottom) coarse modes.

2D cantilever cloak – optimization setup



Beam, loading, and materials

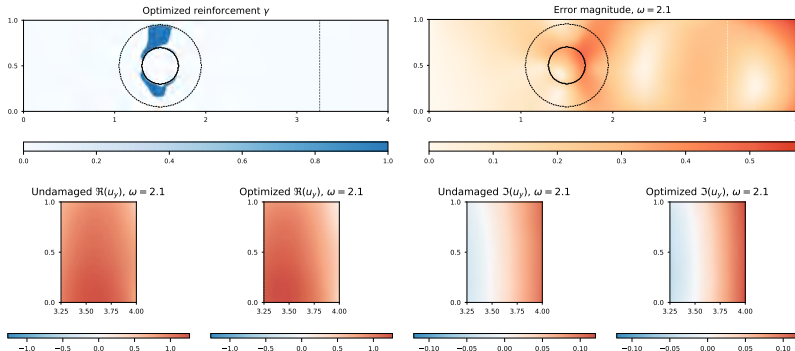
- $[0, 4] \times [0, 1]$ cantilever; combined axial/transverse tip traction
- Soft disk: center $(1.5, 0.5)$, radius 0.2, and $E_d = 0.1E_0$
- Design annulus: outer radius 0.45; reinforced phase $E_1/E_0 = 2.42$
- Common $\nu = 0.294$; $\rho_0 = \rho_d = 0.4$, $\rho_1 = 0.9$
- Mass-proportional damping $\alpha_M = 0.05$; $\beta_K = 0$

Discretization and optimization

- Observation window: $3.25 \leq x \leq 4$
- $\omega = 2.0, 2.4, 2.8, 3.2$ with trapezoidal weights
- 65,536 quadrilaterals; Q_2 displacement; 112 MPI ranks
- 1,053,700 complex displacement DOFs; 65,536 design DOFs
- Cubic stiffness interpolation; no reinforcement-volume constraint

$$\min_{0 \leq \gamma \leq 1} \sum_j w_j \frac{\|\mathbf{u}(\gamma, \omega_j) - \mathbf{u}_0(\omega_j)\|_{L^2(\Omega_{\text{obs}})}^2}{2\|\mathbf{u}_0(\omega_j)\|_{L^2(\Omega_{\text{obs}})}^2}.$$

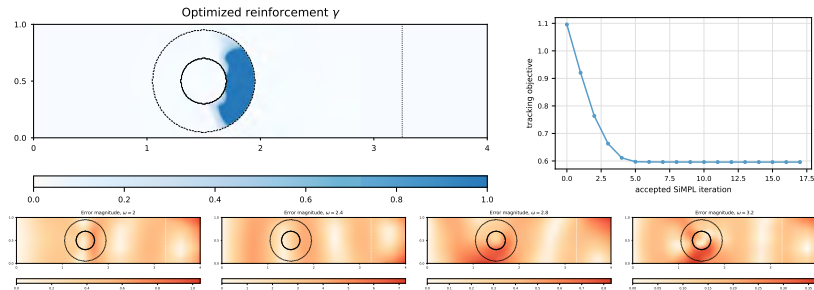
2D cantilever cloak – single-frequency result



Single design frequency $\omega = 2.1$; accepted SiMPL iteration 6:

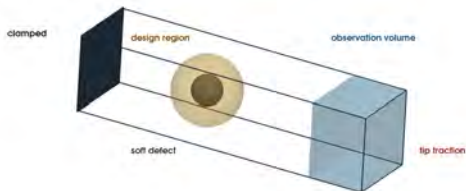
$J = 0.16618 \rightarrow 0.030771$ (81.5% reduction), with final reinforcement fraction 0.19249. Bottom row: transverse displacement in the observation window with common limits for each undamaged/optimized pair.

2D cantilever cloak – optimized design and tracking error



Accepted SiMPL iteration 17: $J = 1.09576 \rightarrow 0.59591$ (45.6% reduction); final reinforcement fraction 0.31649. Solid circle: fixed defect; dashed circle: design boundary; dotted line: start of the observation window.

3D cantilever cloak – problem and solver settings



Physical problem

- Beam $[0, 4] \times [0, 1]^2$, clamped at $x = 0$, with axial/transverse tip traction $(1.25, -1, 0)$
- Fixed soft sphere: center $(1.5, 0.5, 0.5)$, radius 0.2 , $E_d = 0.1E_0$; design radius 0.45 and $E_1/E_0 = 2.42$
- Observation volume $3.25 \leq x \leq 4$; common $\nu = 0.294$; $\rho_0 = \rho_d = 0.4$, $\rho_1 = 0.9$

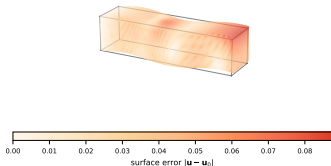
Discrete optimization and solver

- $\omega = 2.0, 2.4, 2.8, 3.2$ with trapezoidal weights; $\alpha_M = 0.05$, $\beta_K = 0$
- 131,072 hexahedra; Q_2 displacement; 560 MPI ranks
- 6,514,950 complex displacement and 131,072 design DOFs; cubic interpolation; no volume constraint
- FGMRES/deflated Split I, 36 modes, assembled CG-AMG for H^{-1}

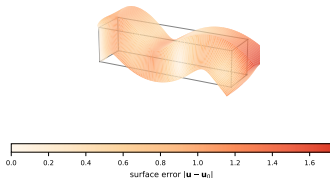
$$\min_{0 \leq \gamma \leq 1} \sum_j w_j \frac{\|\mathbf{u}(\gamma, \omega_j) - \mathbf{u}_0(\omega_j)\|_{L^2(\Omega_{\text{obs}})}^2}{2\|\mathbf{u}_0(\omega_j)\|_{L^2(\Omega_{\text{obs}})}^2}.$$

3D cantilever cloak – deformed optimized response

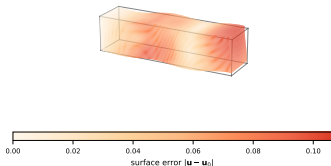
$\omega = 2$: deformed by $\Re(\mathbf{u})$, magnification $\times 0.077$



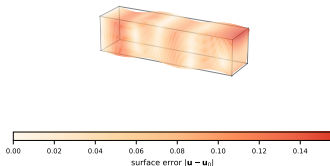
$\omega = 2.4$: deformed by $\Re(\mathbf{u})$, magnification $\times 0.077$



$\omega = 2.8$: deformed by $\Re(\mathbf{u})$, magnification $\times 0.077$

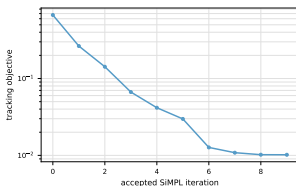
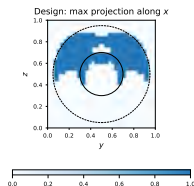
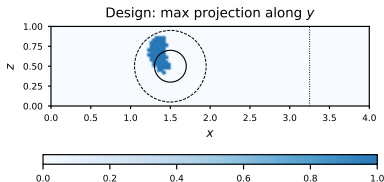
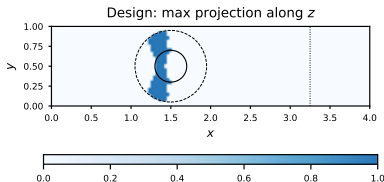


$\omega = 3.2$: deformed by $\Re(\mathbf{u})$, magnification $\times 0.077$



Phase-zero snapshots $\mathbf{x}_{\text{def}} = \mathbf{x} + 0.077 \Re(\mathbf{u})$. Surface color: $|\mathbf{u} - \mathbf{u}_0|$; gray wireframe: undeformed beam.

3D cantilever cloak – optimized design



Accepted SiMPL iteration 9: $J = 0.670523 \rightarrow 0.0101854$ (98.48% reduction); final reinforcement fraction 0.08428. The views show maximum projections of γ along the indicated axis; solid/dashed circles mark the defect/design radii.

Acoustic (scalar) topology optimization

- Scalar complex-pressure Helmholtz model with spatially varying density and bulk modulus
- First-order input/radiation boundaries and a design-independent quadratic sponge layer
- Partial assembly for the state operator; the common solver interface exposes PRESB, splitting, and spectral two-level options
- The coarse space and inner solver must be matched to the acoustic spectrum; low-frequency eigenmodes alone are not uniformly effective in long waveguides
- Application: surround a fixed obstacle by a material distribution that recovers the undisturbed wave downstream

2D acoustic cloak – optimization setup

Waveguide and excitation

- Domain $[0, 15] \times [0, 4]$; unit real incident pressure at $x = 0$
- First-order radiation at $x = 0, 15$; rigid horizontal walls
- Quadratic absorbing sponge on $12 < x < 15$; $\omega = 8$
- Fixed circular obstacle: center $(4, 2)$, radius 0.3
- 300×100 quadrilaterals; Q_2 pressure; 16 MPI ranks

$$\min_{0 \leq \gamma \leq 1} J(\gamma) = \frac{\|p(\gamma) - p_0\|_{L^2(\Omega_{\text{obs}})}^2}{2\|p_0\|_{L^2(\Omega_{\text{obs}})}^2},$$

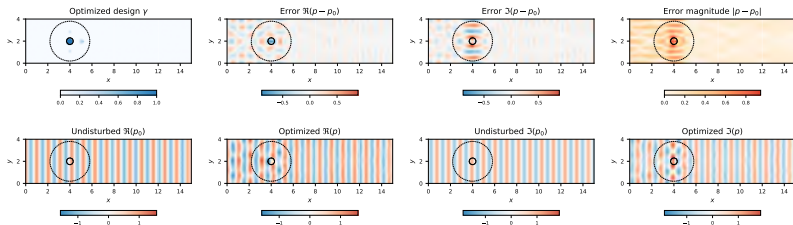
Design and target

- Design annulus:
 $0.3 < |x - (4, 2)| \leq 1.8$
- Observation region:
 $[8, 12] \times [0, 4]$
- Background $(\rho, \kappa) = (1, 1)$; design phase $(0.25, 4)$; fixed obstacle $(25, 25)$
- Cellwise density γ , initialized at zero; no volume constraint
- 120,801 complex-pressure DOFs (241,602 real-block unknowns); 30,000 design DOFs

p_0 : undisturbed pressure.

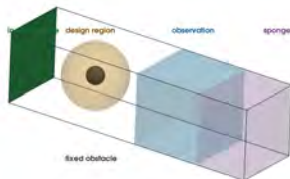
SiMPL: relative gap tolerance 10^{-4} met at accepted iteration 80. Linear solve: FGMRES/PRESB with assembled CG-AMG for H^{-1} .

2D acoustic cloak – optimized design and pressure fields



Accepted SiMPL iteration 80: $J = 4.3931 \times 10^{-2} \rightarrow 4.2021 \times 10^{-3}$ (90.4% reduction). Solid disk: fixed obstacle; dashed circle: outer design boundary; dotted lines: observation window. The error is plotted throughout, but minimized only inside the observation window.

3D acoustic cloak – formulation and geometry



$\Omega = [0, 15] \times [0, 4]^2$, obstacle $B_{0.5}(4, 2, 2)$,
design ball $B_{1.8}(4, 2, 2)$,

$\Omega_{\text{obs}} = [8, 12] \times [0, 4]^2$.

$\omega = 8$; $(\rho, \kappa)_{\text{bg}} = (1, 1)$, $(\rho, \kappa)_{\text{des}} = (0.25, 4)$,
 $(\rho, \kappa)_{\text{obst}} = (25, 25)$.

Mass damping 0.05; sponge amplitude 1; no
volume constraint.

Find the complex pressure $p(\gamma)$
from

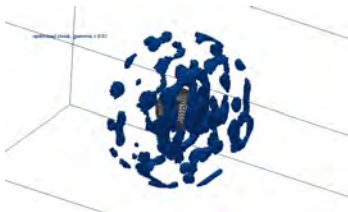
$$a_\gamma(p, v) - \omega^2 m_\gamma(p, v) + i\omega d(p, v) = \ell(v),$$

with unit incident pressure at
 $x = 0$, first-order radiation, rigid
lateral walls, and a quadratic
sponge for $12 < x < 15$. The
design solves

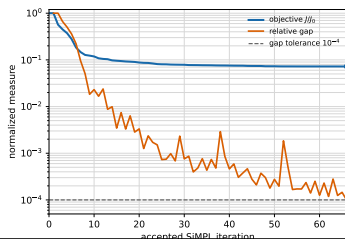
$$\min_{0 \leq \gamma \leq 1} \frac{\|p(\gamma) - p_0\|_{L^2(\Omega_{\text{obs}})}^2}{2\|p_0\|_{L^2(\Omega_{\text{obs}})}^2},$$

where p_0 is the undisturbed
wave. The obstacle and
observation volume are fixed; γ
changes only in the design shell.

3D acoustic cloak – design and convergence



Final cloak ($\gamma > 0.01$); wireframe sphere: fixed obstacle.



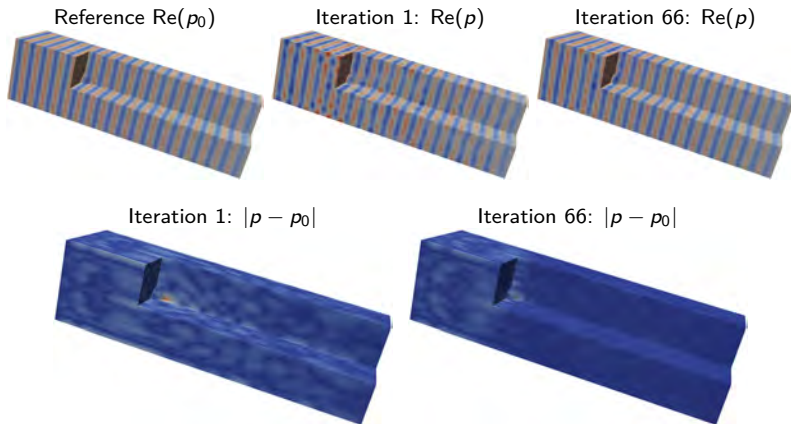
accepted SIMPL iterations	66
$J_0 \rightarrow J_{66}$	$2.5101 \times 10^{-2} \rightarrow 1.8156 \times 10^{-3}$
objective reduction	92.77%
final relative gap	9.934×10^{-5}
final design-shell fraction	4.910×10^{-3}

Relative gap. $g_k^{\text{rel}} = r_k/r_1$, where $d_k = (\psi_k - \psi_{k+1})/\alpha_k$ and

$$r_k = \int_{\Omega_d} [\gamma_k(d_k)_+ + (1 - \gamma_k)(-d_k)_+] , dx.$$

Since r_1 is the reference, $g_1^{\text{rel}} = 1$ and $g_k^{\text{rel}} \rightarrow 0$ at stationarity; stop at $g_k^{\text{rel}} \leq 10^{-4}$.

3D acoustic cloak – recovery of the downstream wave



The initial obstacle strongly scatters the incident plane wave. At iteration 66 the optimized cloak restores the downstream wave and reduces the normalized observation-volume mismatch by **92.77%**.

Summary

- Scalar acoustic and vector elastic models share one frequency-domain polynomial and one MFEM-based solver framework [Anderson et al., 2021, Andrej et al., 2024]
- Modal reduction is a cheap baseline, not a general adjoint-capable solver
- Unsplit PRESB/GMRES (4 its.) and block-diagonal/MINRES (7 its.): mesh- and order-uniform with an exact inverse, but SPD-limited to a finite band
- Indefinite splitting restores positive-semidefinite half-steps, so PRESB stays usable across resonances
- Rayleigh analysis gives the resonance-optimal Method-III parameter; spectral two-level correction roughly halves the outer iterations ($68.4 \rightarrow 37.1$); LOR restores p -robustness
- The 3D elastic cloak reduced the four-frequency tracking objective by **98.48%**
- The 3D acoustic cloak reduced the downstream pressure mismatch by **92.77%** after 66 accepted SiMPL iterations

References |



Anderson, R., Andrej, J., Barker, A., Bramwell, J., Camier, J.-S., Cervený, J., Dobrev, V., Dudouit, Y., Fisher, A., Kolev, T., Pazner, W., Stowell, M., Tomov, V., Akkerman, I., Dahm, J., Medina, D., and Zampini, S. (2021).

MFEM: A modular finite element methods library.
Computers & Mathematics with Applications, 81:42–74.



Andrej, J. et al. (2024).

High-performance finite elements with MFEM.
The International Journal of High Performance Computing Applications, 38(5):450–474.



Axelsson, O. and Karátson, J. (2020).

Superior properties of the PRESB preconditioner for operators on two-by-two block form with square blocks.
Numerische Mathematik, 146(2):335–368.



Axelsson, O., Pourbagher, M., and Salkuyeh, D. K. (2022).

Efficient iteration methods for complex systems with an indefinite matrix term.
Calcolo, 59(2):15.



Axelsson, O. and Salkuyeh, D. K. (2019).

A new version of a preconditioning method for certain two-by-two block matrices with square blocks.
BIT Numerical Mathematics, 59(2):321–342.



Keith, B., Kim, D., Lazarov, B. S., and Surowiec, T. M. (2025).

Analysis of the SiMPL method for density-based topology optimization.
SIAM Journal on Optimization, 35(2):1134–1164.



Kim, D., Lazarov, B. S., Surowiec, T. M., and Keith, B. (2025).

A simple introduction to the SiMPL method for density-based topology optimization.
Structural and Multidisciplinary Optimization, 68:14.

References II



Pazner, W. (2020).

Efficient low-order refined preconditioners for high-order matrix-free continuous and discontinuous Galerkin methods.

SIAM Journal on Scientific Computing, 42(5):A3055–A3083.



Pazner, W., Kolev, T., and Camier, J.-S. (2023).

End-to-end GPU acceleration of low-order-refined preconditioning for high-order finite element discretizations.

The International Journal of High Performance Computing Applications, 37(5):578–599.

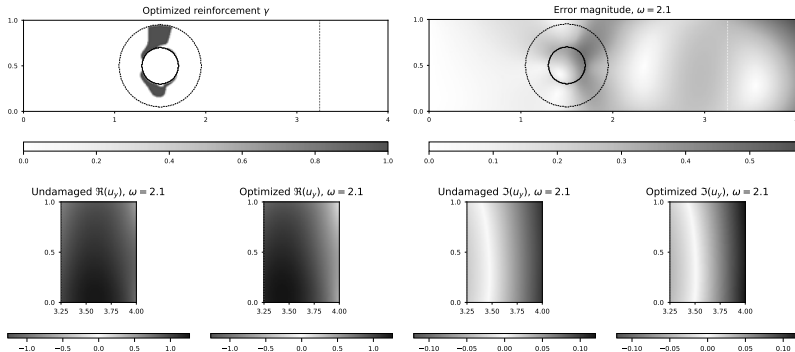
Thank you for your attention. Any questions?



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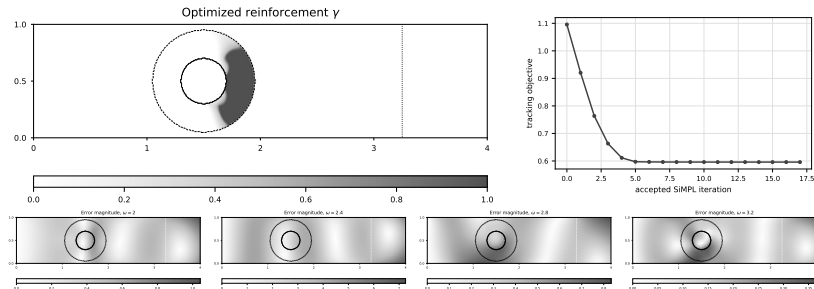
2D cantilever cloak – single-frequency result (grayscale)



Single design frequency $\omega = 2.1$; accepted SiMPL iteration 6:

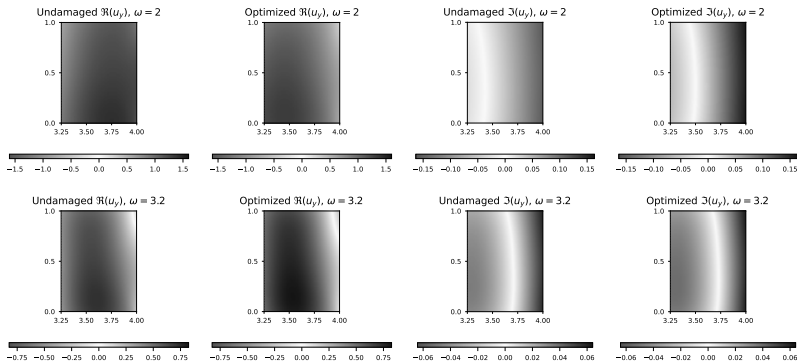
$J = 0.16618 \rightarrow 0.030771$ (81.5% reduction), with final reinforcement fraction 0.19249. Bottom row: transverse displacement in the observation window with common limits for each undamaged/optimized pair.

2D cantilever cloak – design and tracking error (grayscale)



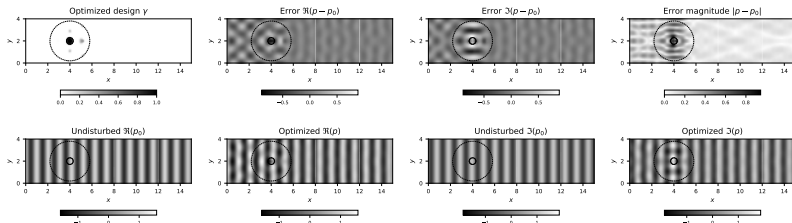
Accepted SiMPL iteration 17: $J = 1.09576 \rightarrow 0.59591$ (45.6% reduction); final reinforcement fraction 0.31649. Solid circle: fixed defect; dashed circle: design boundary; dotted line: start of the observation window.

2D cantilever cloak – tip response (grayscale)



Transverse displacement in the observation window at the ends of the optimization band. Each undamaged/optimized pair uses the same color limits. The intermediate frequencies $\omega = 2.4$ and 2.8 are included in the objective and in the preceding error summary.

2D acoustic cloak – grayscale version



Accepted SiMPL iteration 80: $J = 4.3931 \times 10^{-2} \rightarrow 4.2021 \times 10^{-3}$ (90.4% reduction). Solid disk: fixed obstacle; dashed circle: outer design boundary; dotted lines: observation window. The error is minimized only in the observation window.